CSCI218 Roundup

MCQ

Topic: ML & Computer Vision (6 Questions)

1. [Linear Regression Conditions]
In linear regression with squared error loss, the closed-form solution $w = (X^T X)^{-1} X^T y$ requires which specific linear algebra condition to be computable?
A) X must be a square matrix
B) $X^T X$ must be invertible (full column rank)
C) The training data must be linearly separable
D) The loss function must be convex

2. [Activation Function Gradients]
Why is ReLU ($f(x) = \max(0, x)$) often preferred over sigmoid for hidden layers in deep networks regarding gradient flow?
A) It produces outputs in [0, 1], aiding interpretability
B) It is differentiable everywhere
C) It mitigates vanishing gradients and enables sparse activation
D) It guarantees convex optimization

3. [Loss Function Behavior]
For binary classification with sigmoid output, why is cross-entropy loss preferred over squared error?
A) Cross-entropy is convex for all parameters
B) Cross-entropy provides stronger gradients when predictions are wrong, avoiding saturation
C) Squared error cannot be used with probabilistic outputs
D) Cross-entropy requires fewer training examples

4. [Optimization Noise]
Regarding Stochastic Gradient Descent (SGD) vs. Batch GD, which is TRUE regarding convergence behavior?
A) SGD always converges faster to the global optimum
B) SGD uses one random example per update, introducing noise that can help escape local minima
C) Batch GD is more memory-efficient for large datasets
D) SGD requires convex loss functions

5. [CNN Weight Sharing]
In convolutional layers, using the same filter across spatial locations (weight sharing) primarily provides:
A) Invariance to image rotation
B) Translation equivariance and parameter efficiency
C) Faster convergence during training
D) Better performance on small datasets

6. [Transfer Learning Strategy]

When fine-tuning a pre-trained CNN on a new task with limited data, which strategy is MOST appropriate?

A) Train all layers from scratch with random initialization

B) Freeze all convolutional layers; train only the classifier head

C) Freeze early layers; fine-tune later layers with a small learning rate

D) Use only final fully-connected layer features without modification

Topic: Probabilistic Reasoning (3 Questions)

7. [Decision Theory]

According to Decision Theory, when is an agent considered rational?

A) When it chooses the action that guarantees the best possible outcome.

B) When it chooses the action that yields the highest expected utility, averaged over all possible outcomes.

C) When it chooses the action that minimizes the probability of failure.

D) When it chooses the action that maximizes the probability of the most likely state.

8. [Conditional Independence]

In the Ghostbusters example, why are sensor readings $C_{1,1}$ and $C_{1,2}$ considered conditionally independent?

A) Because they are physically far apart.

B) Because they are absolutely independent variables.

C) Because they are conditionally independent given the Ghost location G.

D) Because the sensor model is uniform across all squares.

9. [Bayes Net Structure]

Consider the Alarm Network Bayes Net (Burglary, Earthquake, Alarm, John, Mary). What does the absence of an arc between Burglary and John imply?

A) Burglary and John are absolutely independent.

B) John never calls during a burglary.

C) John is conditionally independent of Burglary given the Alarm.

D) Burglary causes John to call directly.

Topic: Knowledge & Logic (3 Questions)

10. [Entailment & Models]

Regarding Entailment ($\alpha \models \beta$), which statement about the sets of models is correct?

A) The set of models of α is a superset of the models of β.

B) The set of models of α is disjoint from the models of β.

C) The set of models of α is a subset of the models of β.

D) The set of models of α is identical to the models of β.

11. [Deduction Theorem]

According to the Deduction Theorem, $\alpha \models \beta$ if and only if:

A) The sentence $(\alpha \wedge \beta)$ is valid.

B) The sentence $(\alpha \rightarrow \beta)$ is valid.

C) The sentence $(\alpha \vee \beta)$ is satisfiable.

D) The sentence $(\neg\alpha \rightarrow \beta)$ is unsatisfiable.

12. [Proof by Contradiction]

To prove $\alpha \models \beta$ using Proof by Contradiction, what must be shown?

A) That $(\alpha \wedge \neg\beta)$ is valid.

B) That $(\alpha \wedge \neg\beta)$ is satisfiable.

C) That $(\alpha \wedge \neg\beta)$ is unsatisfiable.

D) That $(\alpha \vee \neg\beta)$ is unsatisfiable.